



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-02 Tool Setup - First Use

Dedicated Python 3.13 environment for dbt Core + DuckDB; Windows-first with WSL-safe boundary

Artifact	C3_029_C3_02_TOOL_SETUP_FIRST_USE_RC1
Status	2026 - LEARNER EDITION
Phase	C3-02 Transformation Engineering & Warehousing 14 lessons 34-46 focused hours
Prerequisite	C3-01 Gate PASS. C2B warehouse/modeling foundations may be assumed; implementation depth is taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: analogy -> tiny example -> guided real tool -> break/fix -> independent changed version -> validation -> production reasoning -> evidence. A successful dbt command alone never proves correct data.



Why C3-02 uses its own Python 3.13 environment

C3-00 taught general Python 3.14 environment habits. The current dbt-duckdb 1.11.0 release advertises Python classifiers through 3.13, so this phase does not assume a C3-00 Python 3.14 venv is compatible. Use a separate `.venv-c3-02`` with Python 3.13. Do not downgrade or alter another course environment.

Locked local teaching runtime (verified 2026-09-10): dbt-core 1.11.15 + dbt-duckdb 1.11.0 + DuckDB 1.5.5 + pytest 8/9 compatible range. dbt v2/Fusion exists, but this local lab intentionally stays on Core v1.x because the DuckDB adapter is the small offline teaching target.

WINDOWS POWERSHELL - RECOMMENDED BEGINNER ROUTE

```
py -0p
py -3.13 --version
py -3.13 -m venv .venv-c3-02
.\.venv-c3-02\Scripts\Activate.ps1
python --version
python -m pip install --upgrade pip
python -m pip install -r requirements-c3-02.txt
dbt --version
python -c "import duckdb; print(duckdb.__version__)"
pytest --version
```

Expected: active Python reports 3.13.x; dbt shows Core 1.11.x and duckdb adapter; DuckDB reports 1.5.5. Save the real output.

IF ``py -3.13`` IS MISSING

Install a current Python 3.13 release from python.org for Windows, then reopen PowerShell and rerun ``py -0p``. Do not replace your C3-00 3.14 installation; multiple interpreters can coexist.



WSL route - only when a Python 3.13 interpreter already exists

Inside WSL run ``python3.13 --version``. If it exists, create ``.venv-c3-02`` there. If it does not, do not add an unofficial PPA just because a lesson told you to. Use the Windows PowerShell route for this local DuckDB phase, or follow your distribution/Python official installation guidance.

```
python3.13 --version
python3.13 -m venv .venv-c3-02
source .venv-c3-02/bin/activate
python -m pip install -r requirements-c3-02.txt
dbt --version
python -c "import duckdb; print(duckdb.__version__)"
```

CREATE THE LOCAL PROFILE

Copy ``profiles.yml.example`` to ``profiles.yml`` inside the starter project. It points to a local `warehouse.duckdb`` file and contains no password. Do not commit real cloud/database credentials.

```
dbt debug --profiles-dir .
dbt parse --profiles-dir .
dbt ls --profiles-dir . --resource-type model
```

Expected: debug says connection test OK; parse succeeds; model names list. If runtime is unavailable, save the exact blocker. Static inspection is useful but cannot count as executed dbt evidence.



First clean run

```
python scripts/static_fixture_audit.py
python scripts/setup_duckdb.py
dbt seed --profiles-dir . # only if your project actually defines seeds
dbt run --profiles-dir .
dbt test --profiles-dir .
```

The supplied project may start with TODO/incomplete models, so `dbt run/test` can fail until the matching lesson is completed. That is intentional. Setup success means the environment/connection works, not that all transformation work is already solved.

RECOVERY ORDER

1) Confirm active Python 3.13 venv. 2) `python --version`. 3) `dbt --version`. 4) `dbt debug --profiles-dir .`. 5) `dbt parse`. 6) inspect first meaningful error. Never delete source data or disable a test as a generic repair.

DO NOT SHIP

Never include `.venv-c3-02/`, `warehouse.duckdb`, `target/`, `logs/`, `.pytest\_cache/`, `\_\_pycache\_\_`, `.pyc`, `profiles.yml` with secrets, or generated evidence inside a canonical starter archive.